Higher order non-linear PDE: Monge's method

The PDE is of the form

$$Rr + Ss + Tt = V \quad \text{--- (1)}$$

where R, S, T and V are functions of x, y, z, p and q.

We know $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$,

$$r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial p}{\partial x}, \quad t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial q}{\partial y}, \quad s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial q}{\partial x}$$

$$s = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial p}{\partial y}$$

We know the total derivatives are

$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy = r\,dx + s\,dy \quad \text{--- (2)}$$

$$dq = \frac{\partial q}{\partial x} dx + \frac{\partial q}{\partial y} dy = s\,dx + t\,dy \quad \text{--- (3)}$$

(2) & (3) give us,

$$r = \frac{dp - s\,dy}{dx}, \quad t = \frac{dq - s\,dx}{dy}$$

$$(1) \Rightarrow R\left(\frac{dp - s\,dy}{dx}\right) + Ss + T\left(\frac{dq - s\,dx}{dy}\right) = V$$

$$\Rightarrow \left(R\,dp\,dy + T\,dq\,dx - V\,dx\,dy\right) - s\left(R\,dy^2 - s\,dx\,dy + T\,dx^2\right) = 0 \quad \text{--- (4)}$$

clearly any relation between x, y, z, p, q which ~~soln~~ satisfies (4) must also satisfy the following two simultaneous eqns

$$R\,dp\,dy + T\,dq\,dx - V\,dx\,dy = 0 \qquad (5)$$

$$R\,dy^2 - S\,dx\,dy + T\,dx^2 = 0 \qquad (6)$$

They are known as Monge's ~~as~~ auxiliary equations.

Eqn (6) ~~being~~ being a quadratic, in general it can be resolved into two equations, say,

$$dy - m_1\,dx = 0 \quad \text{and} \quad dy - m_2\,dx = 0$$

Now two cases arise:

<u>Case 1:</u> m_1 and m_2 are distinct. In this case we combine $dy - m_1\,dx = 0$ with (5), if necessary with $dz = p\,dx + q\,dy$, It will gives us two integrals (say)

$$u_1 = a, \quad v_1 = b$$

These gives $\boxed{u_1 = f(v_1)}$

Similarly for $y - m_2\,dx = 0$:

$$\boxed{u_2 = f(v_2)}$$

Case II: If $m_1 = m_2$, then we get

$$pp + qq = R \quad \text{——— (6)}$$

then we can use Lagrange's method.

Ex: Solve $r = a^2 t$ ——— (1)

We know $r = \dfrac{dp - s\,dy}{dx}$

$$t = \frac{dq - s\,dx}{dy}$$

Putting these values of r and t in (1)

$$\frac{dp - s\,dy}{dx} = a^2 \frac{dq - s\,dx}{dy}$$

$$\Rightarrow (dp\,dy - a^2 dx\,dq) - s\left((dy)^2 - a^2 (dx)^2\right) = 0$$

Hence Monge's A. Eq are

$$(dy)^2 - a^2 (dx)^2 = 0 \quad \text{——— (2)}$$

$$dp\,dy - a^2 dx\,dq = 0 \quad \text{——— (3)}$$

(2) $\Rightarrow$ $(dy + a\,dx)(dy - a\,dx) = 0$

$\Rightarrow dy = -a\,dx$ | $\int dy = \int a\,dx$

$y = ax + C_1$

$\Rightarrow \boxed{y - ax = C_1}$

(3) $\Rightarrow$ $dp\,(a\,dx) - a^2 dx\,dq = 0$

$\Rightarrow \int dp - \int a\,dq = 0 \Rightarrow p - aq = C_2$

$\therefore p - aq = \phi(y - ax)$ ——— (a)

Similarly, $y = -ax$, we get

$$p + aq = \phi_2(y + ax) \quad \text{--- (5)}$$

Solving (4) & (5), we get

$$p = \frac{1}{2}\left[\phi_1(y - ax) + \phi_2(y + ax)\right]$$

$$q = \frac{1}{2a}\left[\phi_2(y + ax) - \phi_1(a - ax)\right]$$

We know

$$dz = p\,dx + q\,dy$$

$$= \frac{1}{2}\left[\phi_1(y - ax) + \phi_2(y + ax)\right] dx$$

$$+ \frac{1}{2a}\left[\phi_2(y + ax) - \phi_1(y - ax)\right] dy$$

$$= \frac{1}{2a}\phi_2(y + ax)(dy + a\,dx) - \frac{1}{2a}\phi_1(y - ax)(dy - a\,dx)$$

integrating,

$$z = \phi_2(y + ax) + \phi_1(y - ax)$$

$$= \psi_1(y + ax) + \psi_2(y - ax)$$

Solve $q^2 r - 2pqs + p^2 t = pq^2$ —— (1)

<u>Ans:</u> We know, $r = \dfrac{dp - s\,dy}{dx}$ and $t = \dfrac{dq - s\,dx}{dy}$

Putting these values of r and s into (1), we get

$$q^2 \frac{dp - s\,dy}{dx} - 2pqs + p^2 \frac{dq - s\,dx}{dy} = pq^2$$

$$\Rightarrow (dp - s\,dy)q^2 dy - 2pqs\,dx\,dy + (dq - s\,dx)p^2 dx = pq^2 dx\,dy$$

$$\Rightarrow -s(q^2 dy^2 + 2pq\,dx\,dy + p^2 dx^2) + q^2 dy\,dp + p^2 dx\,dq - pq^2 dx\,dy = 0$$

$\therefore$ Monge's A.Es are:

$$q^2 dy^2 + 2pq\,dx\,dy + p^2 dx = 0$$

$$\Rightarrow (q\,dy + p\,dx)^2 = 0 \quad \text{—— (2)}$$

and $q^2 dy\,dp + p^2 dx\,dq - pq^2 dx\,dy = 0$ —— (3)

From (2), $q\,dy + p\,dx = 0$

So $dz = p\,dx + q\,dy = 0$,

integrating $z = c_1$

Putting $q\,dy = -p\,dx$ in (3), we get

$$-p\,dx \cdot q \cdot dp + p^2\,dx\,dp + pq\,dx\,dp = 0$$

$$\Rightarrow -q\,dp + p\,dq + pq\,dx = 0$$

$$\Rightarrow \frac{dp}{-p} + \frac{dq}{q} + dx = 0$$

Integrating $-\ln p + \ln q + x = \ln c_2$

$$\Rightarrow \ln p^{-1} + \ln q + \ln e^{x} = \ln c_2$$

$$\Rightarrow \frac{e^{x} q}{p} = c_2$$

$$\Rightarrow e^{x} q = c_2 p$$

$$\boxed{e^{x} q - p f(z) = 0} \Rightarrow p f(z) - e^{x} q = 0$$

So Lagrange's A.E.s are

$$\frac{dx}{f(z)} = \frac{dy}{-e^{x}}, \quad dz = 0 \quad \therefore \boxed{z = c_3}$$

$$\Rightarrow \frac{dx}{f(z)} = \frac{dy}{e^{x}} \Rightarrow \int e^{x}\,dx = \int f(z)\,dy$$

$$\Rightarrow e^{x} + f(z)\,y = c_3$$

$$\Rightarrow$$

$$\Rightarrow y = -\frac{e^{x}}{f(z)} + \frac{c_3}{f(z)}$$

$$\boxed{y = e^{x}\phi_1(z) + \phi_2(z)}$$

Solve $2x^2r - 5xys + 2y^2t + 2(px+qy) = 0$ — (1)

<u>Ans:</u> We have $r = \dfrac{dp - s\,dy}{dx}$, $t = \dfrac{dq - s\,dx}{dy}$

Putting r and t in (1)

$$2x^2 \frac{dp - s\,dy}{dx} - 5xys + 2y^2 \frac{dq - s\,dx}{dy} + 2(px+qy) = 0$$

$$\Rightarrow 2x^2(dp - s\,dy)\,dy - 5xys\,dx\,dy + 2y^2(dq - s\,dx)\,dx + 2(px+qy)\,dx\,dy = 0$$

$$\Rightarrow -s\left(2x^2dy^2 + 5xy\,dx\,dy + 2y^2dx^2\right) + 2x^2\,dp\,dy + 2y^2\,dq\,dx + 2(px+qy)\,dx\,dy = 0$$

Monge's A.Es are:

$$2x^2\,dp\,dy + 2y^2\,dq\,dx + 2(px+qy)\,dx\,dy = 0 \quad \text{— (2)}$$

$$2x^2dy^2 + 5xy\,dx\,dy + 2y^2dx^2 = 0 \quad \text{— (3)}$$

<u>From (3)</u> $\Rightarrow$ $2x^2dy^2 + 4xy\,dx\,dy + xy\,dx\,dy + 2y^2dx^2 = 0$

$$\Rightarrow 2x\,dy(x\,dy + 2y\,dx) + y\,dx(x\,dy + 2y\,dx) = 0$$

$$\Rightarrow (x\,dy + 2y\,dx)(2y\,dx + x\,dy) = 0$$

$\therefore$ $x\,dy + 2y\,dx = 0 \Rightarrow \int \frac{dy}{y} + \int 2\frac{dx}{x} = 0$

$2x\,dy + y\,dx = 0$ $\Rightarrow \ln y + 2\ln x = \ln c_1$

$\Rightarrow \ln(x^2y) = \ln c_1$

$$\boxed{x^2y = c_1}$$

(2) Can be written as

$$2x\,dp\,(x\,dy) + y\,dq\,(2y\,dx) + 2p\,dx\,(x\,dy) + q\,dy\,(2y\,dx) = 0$$

$$\Rightarrow 2x\,dp\,(x\,dy) + y\,dq\,(x\,dy) + 2p\,dx\,(x\,dy) + q\,dy\,(x\,dy) = 0$$

$$\Rightarrow 2x\,dp - y\,dq + 2p\,dx - q\,dy = 0$$

$$\Rightarrow 2(x\,dp + p\,dx) - (y\,dq + q\,dy) = 0$$

Integrating, $2px - qy = c_2$

$\therefore 2px - qy = f_1(xy)$ which is of

the Lagrange's A.E.s are

$$\frac{dx}{x} = \frac{dy}{-y} = \frac{dz}{f_1(xy)} = \frac{dz}{f(c_1)}$$

$$\int \frac{dx}{x} + \int \frac{2\,dy}{y} = 0$$

$$\Rightarrow \ln x + 2\ln y = \ln c_3$$

$$xy^2 = c_3$$

$$\therefore x = \frac{c_3}{y^2}$$

$$\frac{dy}{-y} = \frac{dz}{f(c_1)}$$

$$dz = -\frac{1}{y} f\left(\frac{c_3^2}{y^3}\right) dy$$

$$= \left(\frac{y^4}{3y}\right) f\left(\frac{c_3^2}{y^3}\right) d\left(\frac{1}{y^3}\right)$$

Since $d\left(\frac{1}{y^3}\right) = -\frac{3\,dy}{y^4}$

Integrating, $z = -\frac{1}{3}$

$$dz = \frac{1}{3}\sqrt{}$$

$$dz = -\frac{1}{y} f(c_1)\, dy$$

$$\therefore\ z = -f(c_1)\ln y + c_3$$

$$\therefore\ z = \phi_1(x^2y^2)\ln y + \phi_2(xy)$$

Solve $x^2r - 2xs + t + q = 0$ — (1)

Ans: Monge's Subsidiary of A.Es are

$$x^2\,dp\,dy + dq\,dx + q\,dx\,dy = 0 \quad \text{— (1)}$$

and $x^2(dy)^2 + 2x\,dx\,dy + (dx)^2 = 0$ — (2)

(2) $\Rightarrow$ $(x\,dy + dx)^2 = 0 \Rightarrow x\,dy + dx = 0$ — (3)

$$\Rightarrow \int\frac{dx}{x} + \int\frac{dy}{1} = 0 \Rightarrow \ln xy = \ln c_1$$

$$\boxed{xy = c_1}$$

$$\boxed{\ln x + y = c_1} \quad \text{— (4)}$$

Using (3) into (1),

$$x^2\,dp\,dy + dq\,dx(-x\,dy) + q(-x\,dy)\,dy = 0$$

$$\Rightarrow x\,dp - dq - q\,dy = 0$$

$$\Rightarrow x\,dp - dq + q\frac{dx}{x} = 0 \quad \text{Ans } \boxed{x\,dy = -dx}$$

$$\Rightarrow dp - \frac{dq}{x} - q\frac{dx}{x} = 0$$

$$\Rightarrow \int d\left(p - \frac{q}{x}\right) = 0$$

$$\therefore\ p - \frac{q}{x} = c_2 \quad \text{— (5)}$$

From (4) & (5), the intermidiate integral of the given eqn is

$$p - \frac{q}{x} - \phi_1(y + \ln x) = 0$$

$$\Rightarrow xp - q = x\phi_1(y + \ln x) = 0$$

Lagrange's A.E, $\frac{dx}{x} = \frac{dy}{-1} = \frac{dz}{x\phi_1(c_1)}$

$$\frac{dx}{x} = \frac{dy}{-1}$$

$$\Rightarrow \boxed{\ln x + y = c_3}$$

$$\frac{dx}{x} = \frac{dz}{x\phi_1(c_1)}$$

$$\Rightarrow \int dz - \int \phi_1(c_1)\, dx = 0$$

$$z - x\phi_1(c_3) = c_4$$

$$\therefore \quad z - x\phi_1(y + \ln x) = c_4$$

$$z - x\phi_1(y + \ln x) = \phi_2(y + \ln x)$$